

Homework 1: Classification Models for Water Quality and Potability

Background

Dataset: **Water Quality and Potability**

(<https://www.kaggle.com/datasets/uom190346a/water-quality-and-potability>)

Use the Water Quality and Potability dataset to build classification models that predict whether a water sample is potable (1) or not potable (0).

Water Quality and Potability

Dataset will help to determine whether the water is potable or not



[Data Card](#) [Code \(62\)](#) [Discussion \(1\)](#) [Suggestions \(0\)](#)

About Dataset

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This dataset contains water quality measurements and assessments related to potability, which is the suitability of water for human consumption. The dataset's primary objective is to provide insights into water quality parameters and assist in determining whether the water is potable or not. Each row in the dataset represents a water sample with specific attributes, and the "Potability" column indicates whether the water is suitable for consumption.

Usability ⓘ

10.00

License

[Apache 2.0](#)

Expected update frequency

Never

Tags

[Classification](#)

[Biology](#)

Homework 1: Classification Models for Water Quality and Potability

Your Tasks

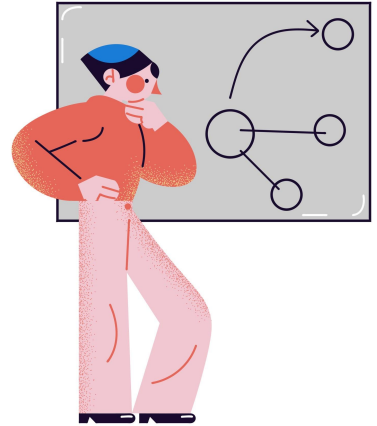
Build and evaluate the following 5 classification models:

1. Decision Tree
2. Naive Bayes
3. Logistic Regression
4. SVM
5. KNN

Test set

For Each Model

1. Train the classification model
2. Make predictions
3. Evaluate the model using:
 - Confusion Matrix
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Classification Report
 - ROC-AUC (if any)



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Data Preprocessing

1. Import the dataset
2. Check for NULL values and remove records containing NULL values using `dropna()`

```
df = df.dropna()
```

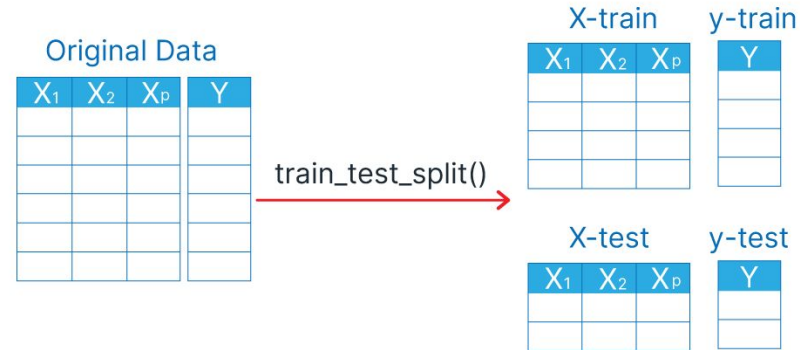
3. Check basic descriptive statistics using `.describe()`
4. Define the input features (X) and target (y)
5. Split the dataset into training and testing sets (80:20)
6. Perform other appropriate data preprocessing, if needed
7. Apply feature scaling when appropriate

Homework 1: Classification Models for Water Quality and Potability

Train/Test split (80:20)

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```



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Compare the Models

Create a summary table comparing the performance of all five models.

Then answer:

1. Which model performs best, and why?
2. Which evaluation metric is most important for water potability classification? Explain your reasoning.

Model	Accuracy	Precision	Recall	F1-score
Decision Tree				
Naive Bayes				
Logistic Regression				
SVM				
KNN				

Important: Explain your results rather than simply reporting the numbers.